

Nerd Heaven

Tech support workshop

Version 1.0 · October 3, 2026

Learn the tools, work through support problems, and build a record of what you can do. Read each lesson, perform its lab, and explain the result. The checkpoints include answers for self-review.

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Know the machine in front of you.
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Build evidence you can stand behind.

Use this with the offline course

Open index.html from the Nerd-Heaven folder for the network, cable, sharing, and ticket benches. Use fictional data and a computer you have permission to practice on. Optional official reading needs internet; the core lessons do not.

Suggested rhythm: review a skill, learn one concept, demonstrate it, then record what happened. Extend the plan to suit your pace.

Name: _____ Started: _____

Start where you are

Task: Build a practice space and choose your starting point.

Technical support is a learned craft. You do not need to memorize every menu or know every answer. You need to notice details, ask useful questions, protect someone else's work, and follow evidence.

This course begins with Windows 11 and everyday office support. Read a lesson, do its lab, answer its checkpoint, and keep a short record. A lesson is complete when you can explain and demonstrate the skill—not when you have merely reached the bottom.

Your practice space

On a computer you are allowed to use, create a folder called **Support-Practice** in Documents. Inside it create **Tickets**, **Files**, and **Evidence**. Use made-up names and data. Keep experiments out of someone else's business files.

The course itself needs only a browser with JavaScript. The command, cable, sharing, and ticket exercises all work offline. Real Windows exercises need a Windows computer. Cloud exercises need internet and an appropriate account; their offline alternatives are included.

A useful rhythm

Try three sessions a week: 10 minutes revisiting the last skill, 25 minutes learning, and 15 minutes practicing. Take longer or shorter as needed. Repeating a difficult task is practice, not a penalty.

Practice

1. Open the Starting check from the menu. Answer before looking up anything; it gives guidance, not a grade.
2. Create the practice folder, or sketch its layout if you do not have a computer yet.
3. In the lesson notebook, write one thing you already know and one thing you want to learn. Export your progress to a file.

Verification: You can open the course, locate a lesson, and explain how to preserve progress when changing computers.

Checkpoint

What makes this lesson complete?

1. Reading every paragraph
2. Being able to explain and demonstrate the skill
3. Finishing faster than someone else

Answer: Being able to explain and demonstrate the skill. A practical demonstration is stronger evidence than time spent reading.

Troubleshoot without guessing

Task: Use a repeatable method and separate symptoms from causes.

A user says, “The internet is broken.” That is a report, not yet a diagnosis. Ask what they were trying to do, what happened instead, the exact error, when it last worked, and who else is affected.

The six-step loop

1. **Understand:** identify the task and business impact.
2. **Scope:** one user, one device, one application, or many?
3. **Observe:** capture the error and current state before changing it.
4. **Hypothesize:** propose a cause and a test that could disprove it.
5. **Change:** with permission, try the smallest reversible fix.
6. **Verify and record:** have the user repeat the original task; document results.

Suppose a website works on another computer but fails in one browser. That narrows the search toward that browser, its session, extensions, or configuration. It does not prove which one. A private window is a useful comparison; it does not reproduce every browser setting and is not a guarantee of anonymity.

Change one variable at a time. Rebooting may restore service, but “fixed after restart” does not identify a root cause. Record what you actually know.

Stop points

Stop and escalate when data loss, suspected compromise, physical danger, privileged changes, or a wider outage exceeds your authority. Good escalation includes evidence; it is not abandoning the user.

Practice

1. Ask a partner to describe a made-up problem vaguely. Ask five clarifying questions without proposing a fix.
2. Write two competing causes and one low-risk test for each.
3. Use Practice desk → No internet. Explain what the evidence rules out and what remains unknown.

Verification: Your proposed next action follows from evidence and has a clear success test.

Checkpoint

A site works in a private window but not a regular window. What is established?

1. The internet provider is down
2. A difference in browser state is relevant
3. The user must reinstall Windows

Answer: A difference in browser state is relevant. The comparison narrows the problem. It does not identify a specific extension, cookie, or policy.

Write a ticket someone can use

Task: Produce a useful handoff and communicate with respect.

A ticket is a record of work, not a diary of frustration. Write so the next technician can continue without repeating your tests.

A complete ticket

Include the affected person and device, symptom and exact error, start time, scope and impact, tests with results, authorized changes, current state, and next action. Record the user's preferred contact method. Never put passwords, recovery codes, or sensitive customer documents in the ticket.

Weak: "Email broken. Restarted. Still bad."

Useful: "One user on laptop NH-014 cannot send from the desktop mail app since 09:10. Web mail sends a test to the agreed recipient successfully. Desktop app reports 'Disconnected.' No other users affected. Checked network and account identity; no configuration changes yet. User can use web mail temporarily. Next: examine desktop app connection state."

Talk like a colleague

Say, "Let's check which account this window is using," instead of "You used the wrong account." Explain before taking control. Ask the user to save work before restarting. Let them type credentials privately.

Prioritize by impact and urgency. A payroll outage affecting ten people may outrank a cosmetic problem for a senior manager. Follow the organization's priority rules rather than inventing a severity label.

A workaround restores a task temporarily. A resolution addresses the incident sufficiently to close it. A root cause explains why it happened. These can be three different things.

Practice

1. Open the Ticket bench. Write a ticket for the desktop-mail example in your own words.
2. Include a user-facing update with the workaround and next step.
3. Export the ticket. Have a partner identify any missing detail before checking the suggested answer.

Verification: Another person can name the scope, completed tests, workaround, and next action from your ticket.

Checkpoint

What is the best closing check?

1. The error vanished once
2. The user can repeat the original task successfully
3. The technician has finished typing

Answer: The user can repeat the original task successfully. Verify the task the user needed, then document the result.

Hardware, software, and the boot path

Task: Identify major components and distinguish power, display, and boot failures.

The processor executes instructions. RAM holds active working data; it is not permanent storage. An SSD or hard drive stores files and the operating system. The motherboard connects components. Firmware initializes hardware before the operating system starts. A dock extends ports but can introduce its own power, display, and compatibility issues.

Before opening anything

Record make, model, serial or asset tag, operating system, and the symptom. A USB-C connector describes shape; it does not guarantee charging, video output, or every data speed. Check the device and dock specifications.

Three different failures

- **No power:** no signs of life. Check a known-working outlet, correct power supply, and visible damage.
- **No display:** the machine may be running. Check monitor power, selected input, cable, and a known-working screen.
- **No operating-system boot:** a logo or error appears, but Windows does not load. Record the message before recovery actions.

Disconnect external devices one at a time only when safe. Do not yank storage devices during writes. Do not open power supplies, handle swollen batteries, or experiment inside equipment while powered. For a damaged battery or electrical smell, stop using the device and follow the site's safety procedure.

For authorized internal work, shut down, unplug, use appropriate electrostatic precautions, and follow the manufacturer's service manual. "Fits the hole" is not a compatibility test.

Practice

1. Inventory your practice computer using Settings → System → About and labels. Do not publish the serial number.
2. Trace power and display cables without disconnecting them. Identify a monitor input and a data port.
3. Write separate first checks for no power, no display, and a Windows boot error.

Verification: You can explain RAM versus storage and avoid treating every dark screen as a dead computer.

Checkpoint

A USB-C plug fits a port. What can you conclude?

1. It supports every USB-C capability
2. The physical connector fits; capabilities still need checking
3. It must carry video

Answer: The physical connector fits; capabilities still need checking. Connector shape alone does not establish power, video, or speed support.

Files, folders, and safe recovery

Task: Use file paths, extensions, copies, and recovery without risking originals.

A path tells you where an item lives. `C:\Users\Alex\Documents\report.docx` describes a local path. `\\fileserver\team\report.docx` is a network share path. A cloud link is a different kind of address. A shortcut points to an item; it is not the item itself.

Build a known-good copy

1. Open File Explorer and go to your practice folder.
2. Use View → Show → File name extensions if available; menu labels may move.
3. Create a plain text file named `practice-original.txt`.
4. Copy it into Evidence. Rename the copy `practice-copy.txt`.
5. Edit only the copy. Compare the contents.

Renaming `.txt` to `.docx` does not convert the file format. An application must perform a real conversion.

Deleted local files may go to the Recycle Bin. Network shares, removable drives, permanent deletion, and cloud storage can behave differently. Never promise recovery without checking. For important lost data, stop writes and escalate; installing recovery tools on the same affected drive can overwrite what you need.

Sync keeps locations aligned and may spread deletion. A backup is a separately recoverable copy with a retention plan. Test a restore with a disposable file before relying on a backup.

Practice

1. Make an original and a copy; change the copy and confirm the original is unchanged.
2. Delete only a disposable local file, then restore it from the Recycle Bin if it appears there.
3. Write where each file lives: local drive, USB, network share, or cloud.

Verification: You can distinguish a copy, shortcut, sync relationship, and backup.

Checkpoint

Changing `photo.jpg` to `photo.pdf` does what?

1. Converts the image into a PDF
2. Changes its name, not its underlying format
3. Backs up the image

Answer: Changes its name, not its underlying format. An extension is a naming clue. Conversion changes the file content through an application.

Slow PCs, updates, and applications

Task: Gather performance evidence and avoid destructive “cleanup.”

“Slow” needs a clock and a task. Does sign-in take three minutes? Does one spreadsheet freeze? Does the browser stall on every site? Ask when the behavior started and what changed.

Windows observation sequence

1. Save work and confirm the exact task that is slow.
2. Open Task Manager with `Ctrl + Shift + Esc`. Inspect Processes and Performance.
3. Note CPU, memory, disk, and network activity during the problem. Sort a column to find heavy users.
4. Check free storage in Settings → System → Storage.
5. Check update status and any requested restart in Settings → Windows Update.

A busy resource is a clue, not automatically a faulty component. High memory use can be normal; sustained pressure plus paging and slow response is more informative. A process name you do not recognize is not sufficient evidence of malware.

Use trusted software sources and approved repair procedures. Do not download “driver boosters,” registry cleaners, or miracle speed tools. Before application repair or reinstall, record version, confirm files are saved, and check how settings and licenses will be preserved.

Windows 10's ordinary support ended in October 2025; special extended-support arrangements exist. For workplace devices, confirm the organization's supported operating systems instead of assuming an old PC is safe because it still starts.

Practice

1. Observe Task Manager with only the browser open, then with a second harmless application. Record what changes.
2. Record free disk space and pending updates without installing or removing anything.
3. Write a ticket describing one timed task and three observations.

Verification: You can collect evidence without killing unfamiliar processes or deleting unknown files.

Checkpoint

CPU usage reaches 100% briefly while opening an app. What next?

1. Replace the CPU immediately
2. Observe duration and whether it matches the symptom
3. Delete every busy process

Answer: Observe duration and whether it matches the symptom. A brief spike can be normal. Correlate measurements with the reported behavior.

Optional official reading: [Windows support](#)

Printers, audio, and meetings

Task: Separate device, connection, queue, and application problems.

A printer incident has several layers: physical device, connection, Windows queue or driver, and the application sending the job. Ask whether everyone is affected and whether the printer itself can produce its own status or test page.

Printer check

1. Check power, paper, consumables, and the display's exact error.
2. Identify USB versus network connection and confirm the intended printer name.
3. In Windows Settings, search for Printers & scanners. Open the relevant queue.
4. Look for paused state or a stuck job. Ask before canceling a job—especially someone else's.
5. Try a Windows test page, then a small harmless document from another app.

If the printer's own page works but the Windows page fails, investigate the path from computer to printer. If Windows prints but one app cannot, narrow toward the app or document. Use the manufacturer's trusted driver source and local procedures.

Meeting audio

Input is the microphone; output is the speaker or headset. Check the app's selected devices, operating-system privacy permissions, hardware mute, app mute, and an app test call where available. A dock or monitor can become the selected audio output unexpectedly. Ask the user before starting a test that others might hear.

Never record a real meeting merely to test recording. Use a private practice meeting with permission.

Practice

1. Locate the printer queue and selected microphone/output devices without changing them.
2. If a practice printer is available, print its own page and a Windows test page. Otherwise draw the four-layer test sequence.
3. Role-play a muted microphone incident; distinguish hardware mute from app mute.

Verification: You can identify which layer failed and test with harmless content.

Checkpoint

A Windows test page prints, but a single PDF does not. Best next direction?

1. Replace the network switch
2. Compare another document and application
3. Factory-reset the printer

Answer: Compare another document and application. The working test page establishes a functioning path for at least that test. Narrow toward the document or app.

Optional official reading: [Microsoft printer troubleshooting](#)

What happens when you open a website?

Task: Explain the jobs of switches, routers, Wi-Fi, DNS, and DHCP.

A computer reaches a website through several cooperating systems. The network adapter connects by cable or radio. A switch moves traffic within a local network. An access point connects wireless devices to that network. A router moves traffic between networks. Home equipment often combines these jobs in one box.

DHCP supplies network configuration such as an address, mask, gateway, and DNS servers. **DNS** translates names into address information. The **default gateway** is the next hop for destinations outside the local subnet. The internet provider carries traffic beyond the site.

A practical path

Device → Wi-Fi access point or switch → router → provider → remote service. DNS helps find the destination; HTTPS protects the application connection using TLS. A successful Wi-Fi connection proves radio association, not that DNS or the website works.

MAC addresses identify network interfaces at the local link layer. IP addresses help route traffic. A device can have several interfaces and addresses, especially with a VPN. A VPN creates a protected connection to another network; it may change routing and DNS.

Ports identify services at the transport layer: HTTPS commonly uses TCP 443, DNS commonly uses UDP or TCP 53. These numbers are clues, not a reason to open firewall ports without approval.

Practice

1. Draw the path from your computer to a website. Label which device performs each job.
2. Explain DNS and DHCP aloud without using their acronym as the explanation.
3. Find the active connection in Windows Settings; note Ethernet or Wi-Fi, not the Wi-Fi password.

Verification: You can explain why “connected to Wi-Fi” and “can use the internet” are different.

Checkpoint

Which service normally provides automatic IP configuration?

1. DNS
2. DHCP
3. HDMI

Answer: DHCP. DHCP supplies configuration; DNS resolves names.

Addresses and subnets without the fog

Task: Read a simple IPv4 configuration and recognize useful warning signs.

IPv4 addresses have four numbers, each from 0 to 255. In the training network `192.168.10.25/24`, the `/24` prefix leaves the first three octets as the network portion. The corresponding mask is `255.255.255.0`.

In this particular subnet, ordinary host addresses are `192.168.10.1` through `192.168.10.254`. `.0` identifies the network and `.255` is its broadcast address. Those last-octet rules are specific to this `/24` example; other prefix lengths divide addresses differently.

A gateway of `192.168.10.1` is on that subnet. A destination `192.168.20.50` is outside it, so routing is needed. Private IPv4 ranges include `10.0.0.0/8`, `172.16.0.0/12`, and `192.168.0.0/16`; they are not normally routed directly on the public internet.

Clues, not verdicts

A Windows IPv4 address beginning `169.254` often means automatic DHCP configuration did not succeed. It supports local-link communication and does not, by itself, provide normal routed IPv4 internet access. IPv6 may still be operating, so examine the active adapter and both protocols before declaring the device completely offline.

A missing gateway, unexpected subnet, duplicate-address warning, or unfamiliar DNS server deserves investigation. Do not invent a static address to “fix” it: you could create a conflict.

Practice

1. On paper, list whether `192.168.10.70` and `192.168.20.70` are local to `192.168.10.25/24`.
2. Run the offline network simulator in DHCP trouble mode. Identify the address clue.
3. With permission, run `ipconfig` on your own Windows PC and label address, mask, and gateway.

Verification: You can interpret the `/24` example and explain why an address alone does not prove a healthy connection.

Checkpoint

On `192.168.10.25/24`, which destination is in the same subnet?

1. `192.168.20.25`
2. `192.168.10.70`
3. `10.0.0.25`

Answer: `192.168.10.70`. For this `/24`, the first three octets must match.

Optional official reading: [Microsoft ipconfig reference](#)

Four commands, useful evidence

Task: Read diagnostic output and avoid overclaiming what a test proves.

Open Command Prompt from Windows Search. These observations normally do not require administrator rights. Begin in the offline simulator if commands are unfamiliar.

Command	Question it helps answer
<code>ipconfig /all</code>	Which adapter, address, gateway, DHCP and DNS settings are in use?
<code>ping 192.168.10.1</code>	Does this example gateway respond to ICMP echo? On a real PC use its actual gateway.
<code>nslookup example.com</code>	Can the configured resolver answer a name query?
<code>tracert example.com</code>	What responding hops can be observed along the route?

A ping reply demonstrates a response to that probe. It does not prove that a website or login service works. A timeout can reflect filtering rather than a dead host. Traceroute stars have the same limitation.

If gateway access works and a name lookup fails, investigate DNS and policy before replacing cables. If a name resolves but HTTPS fails, test the application path and compare scope. Corporate proxies and VPNs can change this sequence.

Record command, target, time, and relevant output. Redact internal names and addresses before sharing externally. Commands such as `release/renew`, `flushdns`, and `network reset` change state; learn why and get approval before using them on work equipment.

Practice

1. Try every supported command in the simulator's healthy mode.
2. Switch to DNS trouble. Compare `ipconfig`, `ping`, and `nslookup` results and write a hypothesis.
3. Explain one thing a successful ping cannot prove.

Verification: You can choose a command to answer a question and state its limits.

Checkpoint

A gateway ping times out. What is proven?

1. The gateway is powered off
2. No reply was received; filtering or other causes remain possible
3. DNS is definitely broken

Answer: No reply was received; filtering or other causes remain possible. A timeout is an observation, not a complete diagnosis.

Optional official reading: [Microsoft nslookup reference](#) · [Microsoft Wi-Fi troubleshooting](#)

Wi-Fi, outages, and safe escalation

Task: Distinguish signal, authentication, routing, and service failures.

Ask whether the device sees the expected network, can join it, receives configuration, and can reach the needed service. These are separate steps. Record location: a problem in one room may be coverage or interference; many users failing at once may be an infrastructure or service incident.

Useful comparisons

- Same device, different authorized connection: Wi-Fi versus Ethernet.
- Same connection, different device: is the fault limited to one endpoint?
- Same device, different application: is one service affected?
- Same app in a browser: does its web version work?

Use only authorized networks. Do not connect a business device to a personal hotspot as a workaround unless policy permits it. Do not disable security software to see what happens.

The 2.4 GHz band often reaches farther but is crowded; higher-frequency bands can offer more capacity with different coverage characteristics. Band support, walls, interference, access-point placement, and client capabilities all matter. Signal bars are an incomplete measure.

When many people are affected

Check the incident channel or service-status page if internet is available. Report scope, start time, locations, symptoms, and comparisons. Do not power-cycle a shared router or switch without authorization: phones, payments, cameras, and other users may depend on it.

Practice

1. Draw a test matrix with two devices and two approved connections. Predict what each pattern suggests.
2. Use Practice desk → Whole office outage. Pick the first responsible action.
3. Write an escalation with scope and three observations, without claiming a cause you have not proven.

Verification: You can separate a single endpoint issue from a site-wide incident.

Checkpoint

Everyone loses connectivity together. Your first move?

1. Reset the router without asking
2. Confirm scope and check/report the shared incident
3. Reinstall every browser

Answer: Confirm scope and check/report the shared incident. A shared failure calls for coordinated infrastructure investigation, not independent disruptive fixes.

Ethernet cable anatomy

Task: Recognize cable components, categories, and installation limits.

A common copper Ethernet cable contains four twisted pairs. Twists help control interference and crosstalk. The eight-position modular connector is commonly called RJ45; you may also hear 8P8C. Patch leads connect devices to wall outlets and patch panels to switches.

A permanent link is the fixed cabling between terminations. A channel includes that link and the patch cords. For ordinary standards-based balanced copper Ethernet installations, plan around a 90 m permanent link and 100 m total channel, subject to cable, application, and environmental requirements. Special extended-reach designs need their own specifications.

Category describes performance requirements, not merely jacket appearance. Cat 5e is commonly used for gigabit Ethernet; Cat 6 and Cat 6A support additional applications under their specified conditions. Do not promise a speed from a category label without considering length and the complete installation.

PoE means Power over Ethernet: compatible equipment can deliver power over the cable as well as data. Never assume a cable is unpowered. Use compatible testers and follow equipment guidance.

Good physical habits

Label both ends, avoid crushing and tight bends, keep required separation from power cabling, use the required fire-rated cable for the space, and do not create trip hazards. Solid-core horizontal cable typically terminates in suitable jacks; stranded patch cable needs compatible plugs. Use the connector manufacturer's instructions.

Practice

1. Identify an Ethernet patch lead and count its eight contacts if visible.
2. Sketch PC → patch cord → wall jack → permanent cable → patch panel → patch cord → switch.
3. Calculate total channel length for 86 m fixed cable plus two 4 m leads.

Verification: You can distinguish permanent link from channel and inspect a cable without assuming it is safe to cut.

Checkpoint

86 m fixed cable plus two 4 m leads totals?

1. 90 m
2. 94 m
3. 104 m

Answer: 94 m. $86 + 4 + 4 = 94$ m. Other installation requirements still apply.

Optional official reading: Fluke: permanent links and channels

T568A, T568B, and making a connection

Task: Use pin numbers and preserve pairs during termination.

T568A and T568B are two standard pin assignments. Neither is intrinsically faster. Follow the site's chosen scheme. A straight-through patch cable uses the same scheme at both ends.

Pin	T568A	T568B
1	White/green	White/orange
2	Green	Orange
3	White/orange	White/green
4	Blue	Blue
5	White/blue	White/blue
6	Orange	Green
7	White/brown	White/brown
8	Brown	Brown

For a modular plug viewed with the gold contacts facing you, the latch behind it, and the cable entering downward, pin 1 is at the left. Verify the connector maker's diagram. A jack's physical color slots need not run in numeric order—follow its labeled A or B layout.

Practice termination sequence

1. Use disconnected practice cable and connectors matched to cable type and conductor size.
2. Strip the jacket to the maker's specified length without nicking conductors.
3. Keep pair twists as close to the termination as the design permits.
4. Arrange or seat wires in the selected scheme, inspect, and terminate with the correct tool.
5. Ensure the jacket is secured for strain relief; test and label both ends.

Do not improvise on a live production run. Fiber requires different tools and handling: never look into a fiber end, and treat shards as a hazard.

Practice

1. Use the Cable bench to arrange all eight T568B positions without the reference, then check.
2. Repeat with T568A and identify which pairs moved.
3. If tools are available, make or punch down a disconnected practice cable under experienced supervision and test it. Otherwise draw and label both ends.

Verification: You can state the viewing orientation, select the same scheme at both ends, and explain why untwisting matters.

Checkpoint

For a straight-through cable, use which schemes?

1. A at one end, B at the other
2. The same selected scheme at both ends
3. Any color order that looks tidy

Answer: The same selected scheme at both ends. A-to-A or B-to-B preserves the chosen straight-through assignment.

Optional official reading: Fluke: T568A and T568B

Testing, patching, and documenting

Task: Interpret wire-map faults and know what a basic tester cannot prove.

A wire-map tester checks how pins connect between ends. An **open** means a connection is missing. A **short** means conductors are electrically joined where they should not be. A **miswire** maps a conductor to the wrong pin.

A **split pair** can have correct numeric continuity but use conductors from different physical pairs. A simple 1-to-1 continuity tester may miss it. A basic green-light result does not certify category performance, usable length, crosstalk, or support for a particular application. Certification requires a suitable instrument and test standard.

Safe port tracing

1. Read labels and consult the site diagram.
2. Record existing patch-panel and switch-port connections.
3. Use a compatible tracer or tester under the site's procedure. Check whether the circuit is live or PoE.
4. Make one approved change and test the endpoint.
5. Update the record and label clearly.

A known-good patch lead is an excellent comparison. If replacing it restores a connection, retest or inspect the original before concluding the entire wall run is faulty.

Do not unplug mystery cables to discover what they do. Do not join switch ports together experimentally: an accidental loop can affect a whole network. Keep a simple record: outlet label, patch-panel port, switch and port, device, test result, date.

Practice

1. Explain why "1 through 8 lights in order" is not the same as certification.
2. Create a five-row patch record for an imaginary office.
3. Use Practice desk → Cable replacement and choose a controlled next test.

Verification: You can describe open, short, miswire, and split pair without overstating a tester result.

Checkpoint

A basic continuity tester passes. What remains unproven?

1. The cable exists
2. Its full category performance and freedom from split pairs
3. There are two ends

Answer: Its full category performance and freedom from split pairs. Some basic testers cannot detect split pairs or measure transmission performance.

Accounts, tenants, and sign-in

Task: Distinguish personal accounts, organizational accounts, licensing, and permissions.

“Office 365” still appears in conversation and some product names. Microsoft 365 combines services and applications, with features depending on the subscription. Word, Excel, and Outlook are apps; Exchange Online, SharePoint, and other services have their own roles.

An organization uses a **tenant**: its administrative environment. A personal Microsoft account and a work or school account are different identities, even when their sign-in names look alike. A browser can be signed into the wrong identity while appearing perfectly functional.

Sign-in triage

1. Ask which service and organization the user needs.
2. Confirm the visible account and exact error. Let the user enter credentials privately.
3. Check connectivity and device date/time.
4. Try the organization's approved web sign-in in a separate browser profile or private window.
5. Record whether authentication succeeds and whether the needed app or file is then accessible.

Authentication proves identity. Authorization grants access. A license enables certain services or app rights. Passing one check does not establish the other two. Administrator roles are another separate layer.

Do not use a personal account to work around an organizational block or upload business files into personal storage. For password or MFA resets, follow identity-verification and approval procedures. If you do not have an admin role, capture evidence and escalate.

Practice

1. Make a table with columns: identity, organization, license/service, resource permission, admin role.
2. Use the Sharing bench to test a correct account with insufficient file permission.
3. Optional online: on your own account, locate the visible account name in a Microsoft app without changing it.

Verification: You can explain why successful sign-in does not guarantee file access.

Checkpoint

A user signs in but cannot open a team file. What is still possible?

1. Missing resource permission
2. The password must be wrong
3. A successful login grants every file

Answer: Missing resource permission. Authentication and authorization are separate.

Optional official reading: [Microsoft 365 help and learning](#)

Outlook and email troubleshooting

Task: Compare web and desktop email and read delivery failures.

Email has both a client and a service. Outlook may be new Outlook, classic Outlook, or web Outlook; menus differ. Begin with the exact app and version instead of assuming a familiar interface.

A small delivery test

1. Confirm the signed-in identity, recipient address, and whether sending, receiving, or both fail.
2. Check Outbox and connection state. Look for an exact error or nondelivery report.
3. With permission, send a harmless test through web mail to an agreed test recipient.
4. Compare the same task in the desktop app.
5. For incoming mail, check search, Junk, rules, and the intended folder before declaring messages lost.

If web mail works but desktop mail fails, the evidence points toward the desktop path, not automatically an organization-wide email outage. A nondelivery report can identify a bad address, policy block, or other server response. Preserve its relevant error and time; do not send sensitive message content outside approved channels.

A shared mailbox is normally accessed through authorized users, not through a password everyone shares. Full Access and Send As are distinct permissions. Storage, licensing, and advanced-feature requirements vary; administrators must check current documentation.

Never delete a mail profile or local data file as a first experiment. Some files contain unsynced or locally archived mail. Confirm data location and recovery plan before a repair.

Practice

1. Use Ticket bench to document a web-versus-desktop comparison.
2. Offline: draft a harmless subject and body, a recipient check, and a success criterion.
3. Optional online: send a test to yourself, reply, and find it by subject. Do not test with a real customer address.

Verification: You can separate client behavior, delivery errors, and shared-mailbox permissions.

Checkpoint

Full Access to a shared mailbox automatically proves Send As permission?

1. Yes
2. No; those are separate permissions
3. Only if the user knows its password

Answer: No; those are separate permissions. Opening the mailbox and sending as it are separate capabilities.

Optional official reading: [Shared mailbox guidance](#)

OneDrive, SharePoint, and Teams files

Task: Find where a file lives and diagnose identity, permission, and sync issues.

Think in storage locations. OneDrive is commonly an individual's organizational workspace. SharePoint provides team sites and document libraries. Teams presents collaboration features over other services: standard channel files use SharePoint, while many chat-uploaded files use the sender's OneDrive. Channel types and sharing rules vary.

“I can't open it”

Ask for the link, exact error, account, intended access, and whether another authorized person can open it. Inspect access rather than sending a fresh copy immediately. A copy can create conflicting versions and expose information.

Use named people and the least permission needed when sharing. A link's existence does not mean everyone can access the file. Organization policy can restrict external sharing. Do not broaden a link to “Anyone” as a shortcut.

“My file disappeared”

Confirm account and location. Compare the browser copy with File Explorer. Check the sync application's status and errors. An online-only placeholder needs connectivity before its content is available. A sync error can leave different versions in different places.

Do not delete or unlink folders to clear an error until unsynced files are preserved. Sync can propagate deletion; recovery and retention features do not replace an organization's independent backup plan.

Practice

1. Use the Sharing bench to give a reviewer read-only access and a collaborator edit access.
2. Draw where a personal work file, team document, and chat attachment might live.
3. Optional online: create a disposable file in your own OneDrive, inspect sharing, and compare web and local status if sync is installed.

Verification: You can check identity, access, storage location, and sync separately.

Checkpoint

A cloud icon on a file often indicates what?

1. It is always saved fully on the PC
2. An online-only item may need download before offline use
3. The file is corrupted

Answer: An online-only item may need download before offline use. A visible placeholder is not proof its contents are available offline.

Optional official reading: [OneDrive Files On-Demand](#) · [OneDrive sync troubleshooting](#)

Word, Excel, and useful office work

Task: Create a small support document and a reliable asset list.

You do not need every feature of Word or Excel to help a user. Learn to produce a clear document, preserve a file, and check a small calculation.

A one-page how-to

1. Open Word, Google Docs, or another available word processor.
2. Create a title and a brief purpose statement.
3. Use heading styles for sections, not only bold formatting.
4. Write a numbered procedure with one action per step.
5. Save an editable copy and export a PDF if available. Reopen both to check layout.

Use a copy before changing a user's formatting. Track Changes or suggesting mode can make edits reviewable. If printing differs from the screen, check page size, margins, scaling, fonts, and the printer selection.

A small spreadsheet

Open the included `practice-assets.csv`. Keep the header row. Sort the whole table, not a single column, so rows remain intact. Add a total below the Cost column with `=SUM(E2:E6)`. It should be 890. A formula begins with an equals sign; cell references identify the inputs.

CSV holds plain tabular data. It does not preserve workbook features such as multiple sheets, formatting, and formulas. Save your working spreadsheet in the application's native format. Use a separate CSV export when needed.

Practice

1. Create the one-page "How to write a useful ticket" document.
2. Open the sample CSV, total the costs, and sort by Location while preserving entire rows.
3. Keep editable copies and a PDF or printout as portfolio evidence. Offline without apps: use paper and calculate the same total.

Verification: Your document is usable by another person and the asset rows stay intact after sorting.

Checkpoint

What does CSV preserve reliably?

1. Multiple formatted worksheets
2. Plain rows and fields
3. Every Excel formula and style

Answer: Plain rows and fields. CSV is a plain data interchange format, not a full workbook format.

Optional official reading: [Microsoft 365 app learning](#)

Google accounts, Gmail, and Calendar

Task: Identify the right account, find mail, and make an unambiguous meeting invitation.

A consumer Google account and an organization's managed Workspace account are not interchangeable. Administrators can control services and sharing. Multiple signed-in accounts make mistakes easy: check the account avatar and address before troubleshooting access.

Find before declaring lost

In Gmail, archive removes the Inbox label without deleting the message. Labels can organize a message in multiple ways. Check the right account, search, Spam, Trash, and relevant labels. Search examples: `from:alex@example.com`, `subject:practice`, and `has:attachment`. Combine criteria to narrow a result. Use the included fictional mail exercise if offline.

Filters can automatically label or move messages. Forwarding and unfamiliar rules can also matter in a security incident. Do not delete a suspicious rule before recording and reporting it through the approved process.

Calendar precision

Confirm date, start and end time, time zone, attendees, meeting location or link, and calendar owner. An event can display at different local times for people in different zones. Editing your own calendar entry is not always the same as changing the organizer's event.

Use a private practice event and ask before inviting anyone. A recurring event has single-occurrence and whole-series choices; read the confirmation carefully.

Practice

1. Offline: write search criteria for mail from Alex with an attachment and a practice subject.
2. Draft an invitation with a date, duration, and explicit time zone.
3. Optional online: search your own test messages and make a calendar event with no guests.

Verification: You can distinguish archive from delete and state the intended time zone.

Checkpoint

Archiving a Gmail message normally means?

1. It leaves the Inbox but remains available elsewhere
2. It is permanently destroyed
3. It is sent to every contact

Answer: It leaves the Inbox but remains available elsewhere. Archive removes Inbox placement; it does not mean permanent deletion.

Optional official reading: [Gmail search operators](#)

Drive sharing and team ownership

Task: Choose appropriate access and distinguish My Drive from shared drives.

Google Drive sharing uses roles such as Viewer, Commenter, and Editor. Choose the smallest role that allows the requested task. Named-person access and general link access are different controls. A user may have the link while the account they are using lacks permission.

Access request sequence

1. Ask which account needs access and what task it must perform.
2. Confirm that the requester is entitled to the file.
3. Have the owner or authorized manager inspect the current sharing.
4. Grant the appropriate role if policy permits.
5. Verify from the intended account and record what changed.

In My Drive, individual ownership matters. In a shared drive, files belong to the team rather than an individual, supporting continuity when people leave. Shared-drive availability and management depend on the organization's edition and policies. Membership, direct sharing, folder restrictions, and external-sharing controls can interact; inspect the actual resource.

Do not solve a blocked external share by making the file public. The restriction may be intentional. Do not transfer ownership or move organizational files into personal storage as an experiment.

For a missing file, ask for its name, last known location, owner, link, and last successful access. Check search and Trash within your authority. Recovery windows and administrator options can differ.

Practice

1. Use the Sharing bench: make Alex a viewer, Jordan a commenter, and Casey an editor. Test permitted actions.
2. Write a handoff plan for a team file when its creator leaves.
3. Optional online: create a disposable document, inspect Restricted sharing, and share only with a consenting practice partner if needed.

Verification: You can explain account mismatch, ownership, and role choice without opening access broadly.

Checkpoint

A reviewer only needs to leave comments. Best role?

1. Editor
2. Commenter
3. Owner

Answer: Commenter. Commenter grants the intended task without unnecessary editing or ownership power.

Optional official reading: [Drive sharing guide](#) · [Shared-drive ownership](#)

Docs, Sheets, Meet, and migration basics

Task: Help someone collaborate and preserve fidelity when moving between suites.

Docs, Sheets, and Slides are collaborative editors. The browser, account, resource permission, and connection all affect the experience. Meet adds device selection and meeting permissions.

A controlled collaboration exercise

Create a disposable how-to document. Use headings and numbered steps. Add a comment to one step and resolve it after revision. Inspect version history and restore only in your practice file. Record what happened instead of relying on memory.

In Sheets, open the sample asset data and verify the total. Freeze the header row for navigation. A filter changes which rows are visible; sorting changes order. Check that a filter isn't hiding the "missing" item before recreating it.

Microsoft ↔ Google

Importing or exporting a document is not a promise of perfect equivalence. Complex formatting, fonts, formulas, macros, and embedded objects may differ. Preserve the original, work on a copy, and compare the result. A PDF is useful for fixed presentation, but not a substitute for an editable source.

For Meet audio or camera issues, inspect selected devices and browser/site permissions, then operating-system privacy settings. Test in a private practice session, not a live meeting. Offline editing in Google apps requires prior setup and has limitations; do not promise that a cloud document will work offline merely because it was opened once.

Practice

1. Offline: create a review checklist comparing original and converted documents.
2. Optional online: make a test Doc with a comment and version history, or practice equivalent review features in a local word processor.
3. Use the sample asset data in any spreadsheet and find an item after filtering.

Verification: You can preserve an original and identify at least three migration checks.

Checkpoint

A converted document looks fine on its first page. Is migration verified?

1. Yes, all formats are equivalent
2. No; compare the rest, formulas, objects, and editable behavior
3. Only the filename matters

Answer: No; compare the rest, formulas, objects, and editable behavior. Verification needs to cover the features and content the user actually relies on.

Optional official reading: [Google Workspace learning center](#)

Passwords, MFA, and identity checks

Task: Help with sign-in without weakening security.

People often contact support when locked out or under pressure. That makes identity verification essential. Use the organization's established verification process before resetting passwords, enrolling a new authenticator, or changing recovery details. A familiar voice, displayed caller number, or urgent story is not sufficient by itself.

Use unique passwords and an approved password manager. Multifactor authentication adds a factor beyond a password. Passkeys and security keys can provide phishing-resistant authentication when supported. A legitimate support worker should not ask a user to read out an MFA code or approve a prompt they did not initiate.

A safe recovery conversation

1. Identify the account and the user's intended task.
2. Verify identity through an approved independent method.
3. Use authorized recovery procedures; never bypass MFA for convenience.
4. Let the user enter their new secret privately.
5. Confirm sign-in and document the action without recording secrets.

If someone receives repeated unsolicited MFA prompts, do not tell them to approve one to make the noise stop. Treat the pattern as suspicious and report it. If you lack the role to recover the account, explain the next authorized step clearly.

Do not store real passwords in your course notebook or ticket exports. This course's progress files are plain text and are not encrypted.

Practice

1. Role-play a caller asking for an urgent MFA reset. Write the verification and escalation steps.
2. Identify which information belongs in a ticket and which must never be recorded.
3. Use Practice desk → Unexpected MFA.

Verification: You can help a locked-out user while preserving identity controls.

Checkpoint

An unexpected MFA prompt appears. Best response?

1. Approve it to stop the notifications
2. Do not approve; report through the trusted channel
3. Send the code to whoever calls

Answer: Do not approve; report through the trusted channel. Only approve authentication you intentionally initiated and understand.

Optional official reading: [CISA multifactor guidance](#)

Phishing, backups, and incident response

Task: Recognize suspicious requests and preserve recoverable work.

Phishing tries to make someone act on a false identity or story. Watch for unexpected credential requests, attachments, payment changes, unusual domains, and pressure. Polished language and a padlock do not prove legitimacy. Attackers can use compromised real accounts.

For a payment or account change, verify through a known independent channel—not the phone number or link supplied by the suspicious message. Do not open an attachment “to see if it's malicious.” Report using the organization's mechanism.

If something was clicked

Ask what happened and when without blame. Report promptly, preserve the message and observations, and follow the incident procedure. If compromise is suspected, authorized isolation may limit spread. Do not run random cleanup tools or wipe the machine: investigators may need the state and evidence. A clean-looking screen is not proof the incident is over.

Recovery you can demonstrate

Keep multiple copies, on different media or services, with at least one appropriately separated from the main system. Version history and recycle bins are useful, but retention and account compromise can limit them. Decide what is backed up, how often, how long it is kept, who can restore it, and test recovery.

An offline USB copy should be ejected safely and disconnected after use. Protect sensitive backups with appropriate encryption and access controls. This course uses fictional data for practice.

Practice

1. Write an independent-verification plan for a message demanding changed bank details.
2. Copy a disposable practice file to a second location. Change the working copy, then restore a separate test copy from the backup.
3. Record the backup date, restore steps, and what was verified.

Verification: You can show a restore and explain why sync alone is insufficient.

Checkpoint

A message has perfect grammar and an HTTPS link. Is it trustworthy?

1. Automatically
2. No; identity and request still need verification
3. Only if it mentions your name

Answer: No; identity and request still need verification. Presentation quality and encrypted transport do not establish the sender's legitimacy.

Optional official reading: [NIST phishing guidance](#)

Remote support and AI without blind trust

Task: Use assistance while retaining authorization, evidence, and judgment.

Remote support can be useful when the user knows who is connecting and consents. Use the organization's approved tool. Explain what will be visible, ask before taking control, and disconnect when finished. Never install persistent access or bypass security controls to make a session easier.

Before acting on instructions

Read a proposed command completely. Identify what it reads, changes, deletes, or sends. If you cannot explain its effect, do not run it on a real system. Check trusted documentation, try a disposable lab where appropriate, and ask an experienced technician. Copying a command from a website or AI chat does not make it authorized.

AI can help explain an error, compare hypotheses, draft a ticket, or quiz you. It can also invent menus, commands, and confident diagnoses. Give it redacted or fictional evidence. Never paste passwords, recovery codes, private customer files, or confidential tickets into an unapproved service.

A useful prompt is: "Here is fictional diagnostic output. Suggest three hypotheses, a low-risk test for each, and what each result would mean. Separate observations from assumptions." Then check the answer.

Keep the user informed. An honest "I need to verify that before changing it" is more professional than a confident guess that risks their work.

Practice

1. Draft an AI question using only fictional data from the network simulator.
2. Identify which suggested commands would be observation versus a state-changing action.
3. Role-play requesting permission for a remote session and ending it cleanly.

Verification: You can explain a proposed action before running it and protect confidential information.

Checkpoint

An AI suggests an unfamiliar administrator command. What next?

1. Run it because the answer sounds confident
2. Understand and verify it before any authorized use
3. Send it to the user with no explanation

Answer: Understand and verify it before any authorized use. The technician remains responsible for authorization and effect.

Capstone: support a small office

Task: Handle three incidents and produce a professional handoff.

Your fictional client is a five-person community office. They use Windows laptops, a shared printer, Google Workspace for documents, and Microsoft 365 for a partner's shared resources. You are authorized to observe, help users with their own files, and propose changes. You are not authorized to reset shared infrastructure, broaden organization policies, or recover accounts without verification.

Three incidents

1. One laptop shows a 169.254 IPv4 address and has no routed IPv4 access. A colleague's laptop works on the same Wi-Fi. IPv6 status is unknown.
2. A volunteer cannot edit a Drive document. They can open it and are listed as Viewer. The owner has approved editing for that volunteer's managed account.
3. The shared printer produces its own test page. One user's Windows test page works, but a particular document does not print.

For each, write the symptom, scope, evidence, hypothesis, safe next test, expected result, workaround if available, and escalation boundary. Do not invent results for tests you have not performed. Use the simulator and permission bench where applicable, then label them as simulated.

Deliverables

Export three tickets, a one-page user guide, your asset spreadsheet, and a brief summary of unresolved questions. Ask a partner to play the user and verify your explanations. A clear "unknown" earns more trust than a fabricated fix.

Practice

1. Use the Ticket bench to document each incident separately and export it.
2. For the network incident, run the simulator in DHCP trouble and note the limitation concerning IPv6.
3. Use the capstone rubric in Field kit to review your work with a partner.

Verification: Each handoff includes evidence, a safe next action, a verification test, and an honest limit.

Checkpoint

You have not run a proposed test. What goes in the ticket?

1. A made-up successful result
2. The planned test and its expected outcomes, clearly labeled
3. Nothing until everything is fixed

Answer: The planned test and its expected outcomes, clearly labeled. Distinguish completed work from planned work.

Make your work visible

Task: Assemble credible evidence of practical skill.

A small portfolio can show how you think. It does not need a fancy website. Keep a folder containing your one-page how-to, asset sheet, three anonymized practice tickets, a network diagram, and a cable record. Label every simulated exercise as practice.

Describe the work accurately

Instead of “Expert in networking,” write, “Practiced reading IPv4 configuration, comparing gateway and DNS tests, and documenting a controlled troubleshooting sequence.” Instead of claiming tenant administration you have not done, say, “Studied identity, licensing, and sharing; completed offline permission exercises.”

In an interview, explain a problem, the evidence you gathered, the action you chose, and how you checked the result. Include a time you changed your hypothesis after a test. That shows judgment.

Practice a 90-second answer

“A user could not open a team document. I confirmed the account and exact error, checked the scope, and found their approved account lacked the required role. The owner authorized the appropriate access. We verified the original task and documented the change.” Use an actual practice case and identify it as such.

Never use a real employer's customer information in a portfolio. Remove names, addresses, asset identifiers, screenshots, and secrets unless you have explicit authorization and an appropriate reason.

Practice

1. Create a portfolio index listing your five artifacts and the skill each demonstrates.
2. Practice your explanation aloud with a partner; ask where it was unclear.
3. Write two accurate résumé bullets about work you actually completed.

Verification: Your claims match your evidence and distinguish practice from employment experience.

Checkpoint

Best description of an offline admin exercise?

1. Managed a live enterprise tenant
2. Completed a simulated permissions exercise
3. Certified administrator

Answer: Completed a simulated permissions exercise. Clear scope makes your evidence credible.

Your next month and the first job

Task: Choose a realistic practice and application plan.

Search role descriptions such as help desk technician, service desk analyst, desktop support, field support, and IT support assistant. Compare duties rather than titles alone. Some jobs emphasize phone support, some physical equipment, and some organization-specific apps.

A four-week practice plan

- **Week 1:** support method, Windows observations, file handling, and two tickets.
- **Week 2:** networking commands, address reading, cable bench, and a network diagram.
- **Week 3:** Microsoft and Google identity, mail and file access; make the how-to and spreadsheet.
- **Week 4:** security scenarios, capstone, portfolio review, and interview practice.

Extend the plan as needed. If a lesson's checkpoint is easy, demonstrate the lab and move on. If it is difficult, revisit the glossary and ask a partner to work one example with you.

Certifications can structure learning and help with some hiring processes, but requirements, exams, and prices change. Read current official objectives before paying. This course is a practical foundation, not accredited certification or a guarantee of employment.

Look for supervised opportunities to help with a consenting person's noncritical equipment or a community practice lab. Agree on scope, preserve backups, and do not take responsibility for systems beyond your current authority. Reliability includes keeping appointments, documenting work, and saying when you need help.

Practice

1. Choose three role descriptions and make a skill-gap list without applying or paying yet.
2. Select two portfolio items to improve during the next week.
3. Agree with your mentor on a practice meeting and one skill to demonstrate.

Verification: You have a next step, a practice partner if available, and honest evidence of your current skills.

Checkpoint

What should guide a certification purchase?

1. An old price mentioned in a forum
2. Current official objectives and relevance to target jobs
3. The most impressive logo

Answer: Current official objectives and relevance to target jobs. Match the learning and cost to your goal using current primary information.